Review.

最后讲这个.

$A \in M_{m,n}(\mathbb{R})$, column vectors $c_1, c_2, \cdots, c_n$, row vectors $a_1, \cdots, a_m$, then $Ax = b \Leftrightarrow x_1c_1 + \cdots + x_nc_n = b \Leftrightarrow a_i \cdot x = b_i$.

In particular, $Ax = 0 \Leftrightarrow x$ is orthogonal to every row vector of A.

Important theorem.

row rank = col rank = rank.

$Ax = b$ is consistant $\Leftrightarrow$ rank A = rank $(A|b)$.

If the system is consistant and rank A = r, then

number of free variable = n-r, then

number of free variables = $n - r$.

$n = r$. the solution is unique. $r < n$. there are infinitely many solution.

Q. Is there a linear system with exactly two solutions.

If x_0 is one particular solution of $Ax = b$. then

$$\{x : Ax = b\} = x_0 + \{x : Ax = 0\}.$$

For $A \in M_n(\mathbb{R})$, TFAE. $\det A \neq 0$.

① A is invertible ② $\operatorname{rank} A = n$. ③ $Ax = 0$ has only the zero solution ④ the reduced row echelon form of A is I_n.

⑤ $\forall b \in \mathbb{R}^n$, $Ax = b$ has a unique solution.

$x = A^{-1} b$.

$(AB)C = A(BC)$. $AB \neq BA$ generally.

$(A+B)^T = A^T + B^T$, $(AB)^T = B^T A^T$

$(AB)^{-1} = B^{-1} A^{-1}$, $(A^{-1})^T = (A^T)^{-1}$.

$D = diag(d_1, \cdots, d_n)$. $D^k = diag(d_1^k, \cdots, d_n^k)$.

invertible $\Leftrightarrow$ every $d_i \neq 0$, $D^{-1} = diag(d_1^{-1}, \cdots, d_n^{-1})$

symmetric matrix if $A^T = A$.

skew-symmetric matrix $A^T = -A$.

$$A = \underbrace{\frac{A + A^T}{2}}_{\text{symmetric}} + \underbrace{\frac{A - A^T}{2}}_{\text{skew-symmetric.}}$$

A, B symmetric. AB symmetric $\Leftrightarrow AB = BA$.

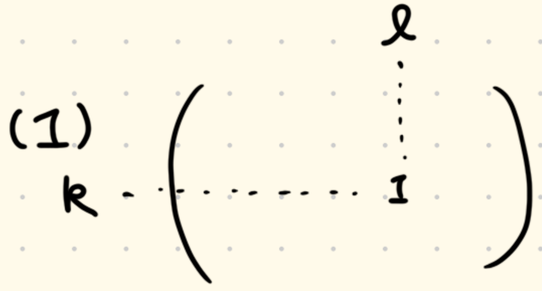
homework review.

9.20 2. (1), (2), (3)

$A = (a_{ij}) \in M_n$. compute (1) $E_{kl} A$ and $A E_{st}$.

(2) $E_{kl} E_{st}$ (3) $tr(E_{lk}^T E_{st})$.

(1)



$E_{kl} A$ has row k equal to row l of A

$$(E_{kl} A)_{ij} = \delta_{ik} a_{lj}.$$

AE_{st} has column t equal to column s of A

$$(AE_{st})_{ij} = \delta_{jt} a_{is}.$$

(2) $E_{kl} E_{st} = \delta_{ls} E_{kt}$.

(3) $tr(E_{lk}^T E_{st}) = tr(E_{kl} E_{st}) = tr(\delta_{ls} E_{kt})$

$= \delta_{ls} \delta_{kt}$.

$A \in M_{m\times n}(\mathbb{R})$, with rows $r_1, \dots, r_m$ and cols $c_1, \dots, c_n$

$T \in M_{s\times m}$,

$S \in M_{n\times t}(\mathbb{R})$.

row k of $TA = \sum_{i=1}^{m} t_{ki} r_i$ is therefore a linear combination of the rows of A.

col l of $AS = \sum_{j=1}^{n} S_{jl} c_j$ is a linear combination of the columns of A.

Left multiplication changes row, right multiplication changes columns.

练习 1.18 (4) $M = (m_1 | m_2)$

M is invertible and $Mx = b$, $Mx' = b'$, then

$M(x+x') = b+b'$. By uniqueness. the solution corresponding

to $b+b'$ is $x+x'$.

Cramer's rule $\rightarrow$ $Ax = b$, $x_i = \det(A_i)/\det(A)$, $i = 1, \cdots, n$

where A_i is the matrix formed by replacing the i-th column of A by the column vector b.

$x_1 = \det(b|m_2)/\det M$, $x_2 = \det(m_1|b)/\det M$.

$\rightsquigarrow \det(b+b'|m_2) = \det(b|m_2) + \det(b'|m_2)$. $\cdots$

Cramer's rule.

$$\begin{cases} 5x_1 + 7x_2 = 3 \\ 2x_1 + 4x_2 = 1 \end{cases} \qquad x = \begin{pmatrix} \frac{5}{6} \\ -\frac{1}{6} \end{pmatrix}.$$

Inverse.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad ad - bc \neq 0.$$

cacculate inverse (row operation. column operation)

$$(A \mid I) \rightsquigarrow (I \mid A^{-1}) \qquad (A \mid B) \rightsquigarrow (I \mid A^{-1}B)$$

$$\left(\frac{A}{I}\right) \rightsquigarrow \left(\frac{I}{A^{-1}}\right) \qquad \left(\frac{A}{B}\right) \rightsquigarrow \left(\frac{I}{BA^{-1}}\right)$$

2.4 1.(6) $\begin{pmatrix} 1 & 2 & 0 & -1 \\ -3 & -5 & 1 & 2 \\ 1 & 3 & 2 & -2 \\ 0 & 2 & 1 & -1 \end{pmatrix}$ $\xrightarrow{\text{inverse}}$ $\begin{pmatrix} -3 & -1 & 1 & -1 \\ -3 & -1 & 0 & 1 \\ -4 & -1 & 1 & 0 \\ -10 & -3 & 1 & 1 \end{pmatrix}$.

$A \in M_{m,n}(\mathbb{R})$, column vectors $c_1, c_2, \cdots, c_n$, row vectors $a_1, \cdots, a_m$, then $Ax = b \Leftrightarrow x_1c_1 + \cdots + x_nc_n = b \Leftrightarrow a_i \cdot x = b_i$.

In particular, $Ax = 0 \Leftrightarrow x$ is orthogonal to every row vector of A.

Inner product space.

orthogonal projection. $a \neq 0$. want to write $u = p + r$.

$p \parallel a$, $p \perp a$. $\rightsquigarrow$ $0 = (u - ta \mid a) = (u \mid a) - t \|a\|^2$.

$$\Rightarrow t = \frac{(u \mid a)}{\|a\|^2}, \quad p = \frac{(u \mid a)}{\|a\|^2} a, \quad r = u - p.$$

$p = Pu$. $P = \dfrac{aa^T}{a^T a}$, prop. $P^2 = P$. $P^T = P$.

$$p = \frac{a^T u}{a^T a} a = \frac{a a^T}{a^T a} u = Pu.$$

$u = (2, -1, 3)^T$, $a = (1, 2, 3)^T$.